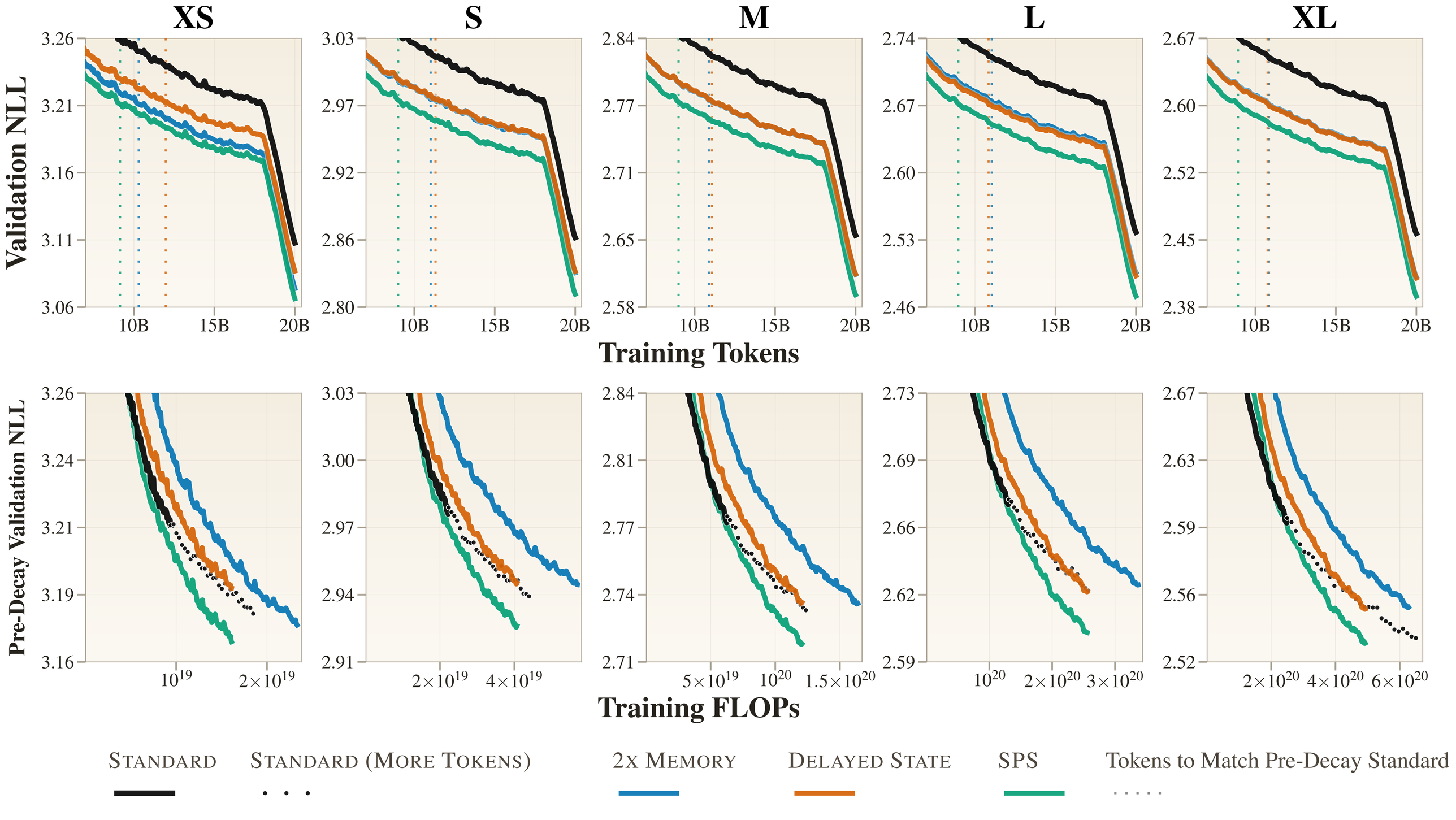

XS
S
M
L
XL
Validation NLL
3.26
3.21
3.16
3.11
3.06
3.03
2.97
2.92
2.86
2.80
2.84
2.77
2.71
2.65
2.58
2.74
2.67
2.60
2.53
2.46
2.67
2.60
2.52
2.45
2.38
10B
15B
20B
Training Tokens
Pre-Decay Validation NLL
3.26
3.24
3.21
3.19
3.16
3.03
3.00
2.97
2.94
2.91
2.84
2.81
2.77
2.74
2.71
2.73
2.69
2.66
2.62
2.59
2.67
2.63
2.59
2.56
2.52
10^{19}
2×10^{19}
4×10^{19}
5×10^{19}
10^{20}
1.5×10^{20}
2×10^{20}
3×10^{20}
4×10^{20}
6×10^{20}
Training FLOPs
Standard
Standard (More Tokens)
2x Memory
Delayed State
SPS
Tokens to Match Pre-Decay Standard